Variety of Jordan algebras and moment map

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SICM-SUSTech

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Jordan algebras arose from the search of "exceptional" setting for quantum mechanics.
In the usual interpretation - the Copenhagen model -

- the physical observables are self-adjoint or Hermitian matrices (or operators on Hilbert space);
- the basic operations are multiplication by $\lambda \in \mathbb{C}$, addition, multiplication of matrices, adjoint operator.

Most of matrix operations are not observable!
In 1932 the German physicist Pascual Jordan proposed a program to discover "a new algebraic setting for quantum mechanics“, free from dependence of matrix structure. Jordan has chosen a new observable operation called **quasi-multiplication**

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Next step is to decide empirically what axioms or laws this operation obeys.

Definition

A **Jordan algebra** consists of a real vector space equipped with a bilinear product $x \cdot y$ satisfying the commutative law and the Jordan identity: $(x^2 \cdot y) \cdot x = x^2 \cdot (y \cdot x)$, $x, y \in J$. A Jordan algebra if **formally real** if

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 0 \quad \Rightarrow x_1 = x_2 = \cdots = x_n = 0.$$

Any associative algebra A over $\mathbb{R}$ gives rise to a Jordan algebra A^+ under quasi-multiplication $x \cdot y = \frac{1}{2}(xy + yx)$.

A Jordan algebra is called **special** if it can be realized as a Jordan subalgebra of some A^+.

If an associative algebra A with involution $\star$, then the subspace of hermitian elements $H(A, \star) = \{x^\star = x \mid x \in A\}$ forms a special Jordan algebra.

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Another example of quasi-multiplication was found by Jordan on $J := \mathbb{R}^n \oplus \mathbb{R}1$ for $n \geq 2$. Namely 1 acts as identity element and for any $u, v \in \mathbb{R}^n$ their product is given by inner product in $\mathbb{R}^n$

$$u \cdot v = \langle u, v \rangle 1.$$

Having settled on the basic axioms for their systems, it remained to find exceptional Jordan algebras Jordan hoped that by studying finite-dimensional algebras he could find families of simple exceptional algebras E_n parametrized by natural numbers n, so that letting n go to infinity would provide a suitable infinite-dimensional exceptional home for quantum mechanics.

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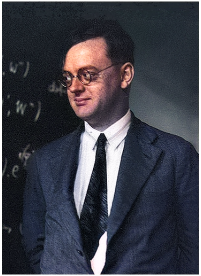

In a fundamental 1934 paper Jordan, Wigner and von Neuman showed that every finite-dimensional formally real Jordan algebra is a direct sum of a finite number of the following algebras

- four types of Hermitian $n \times n$ matrix algebras $H_n(D)$ corresponding to the four real division algebras D (the reals, complexes, quaternions, and the octonions or Cayley algebra $\mathbb{O}$) of dimensions 1, 2, 4, 8 (but for the octonions only $n \leq 3$ is allowed).
- Jordan algebra of Clifford type $J(\mathbb{R}^n)$, $n \geq 2$.

 

Albert showed that $\mathcal{A} = H_3(\mathbb{O})$ is indeed exceptional Jordan algebras of dimension 27 which now is called **Albert algebra**.

Theorem (Zelmanov, 1983)

Any simple Jordan algebra, of arbitrary dimension, is either

1. *an algebra of Hermitian elements $H(A, \star)$ for a $\star$-simple associative algebra with involution, or*
2. *an algebra of Clifford type determined by a nondegenerate symmetric form, or*
3. *an Albert algebra $\mathcal{A}$ of dimension 27 over its center.*

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Back to the main topic of my talk

- Introduce techniques from **Geometric Invariant Theory** (GIT) in the study of finite-dimensional complex Jordan algebras.
- Consider a naturally defined **moment map** for the affine variety of n-dimensional Jordan algebras and the associated **energy functional**, and use it to define a notion of **soliton** Jordan algebra.
- The search of a soliton in a given isomorphism class of Jordan algebras can also be thought of as the search for the "best"(Hermitian) metric in a given Jordan algebra.
- Relate the new geometric invariants to "old"algebraic invariants of Jordan algebras.
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Varieties of associative, Lie and Jordan algebras

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A **Jordan algebra** is a commutative $\mathbb{C}$-algebra J satisfying

$$(x^2y)x = x^2(yx) \quad x, y \in J.$$

We can view a commutative algebra of dimension n as a tensor

$$\mu \in S^2(\mathbb{C})^* \otimes \mathbb{C}^n \simeq S^2(\mathbb{C}^*) \otimes \mathbb{C}^n =: V_n$$

Fix a basis $\{e_1, \ldots, e_n\}$ of V. To define an algebraic structure

$$e_i \cdot e_j = \sum_{k=1}^{n} \mu_{ij}^k e_k, \text{ for all } i, j \in \{1, \ldots, n\}.$$

The Jordan identity provides polynomial equations which cut out the affine variety Jor_n in V_n of n-dimensional Jordan algebras.

$$\sum_{a=1}^{n} \mu_{ij}^a \sum_{b=1}^{n} \mu_{kl}^b \mu_{ab}^p - \sum_{a=1}^{n} \mu_{kl}^a \sum_{b=1}^{n} \mu_{ja}^b \mu_{ib}^p + \sum_{a=1}^{n} \mu_{lj}^a \sum_{b=1}^{n} \mu_{ki}^b \mu_{ab}^p -$$

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for all $i, j, k, l, p \in \{1, \ldots, n\}$.

A point $(\mu_{ij}^k) \in Jor_n$ represents an n-dimensional $\mathbb{C}$-algebra along with a particular choice of basis. A change of basis in μ gives rise to a possible different point of Jor_n, or equivalently there is a natural action of $G := GL(n, \mathbb{C})$ on V_n:

$$g \cdot \mu(x, y) = g(\mu(g^{-1}x, g^{-1}y)), \quad x, y \in \mathbb{C}. \tag{1}$$

The orbit $G \cdot \mu$ yields the isomorphism class of μ and Jor_n is invariant subvariety.

For $\mu, \nu \in Jor_n$ we call ν a **deformation** of μ if $G \cdot \nu$ contains μ in its Zariski closure (which is the same as closure in the Hausdorff topology) that is, $\mu \in \overline{G \cdot \nu}$. In this case, we also say that μ is **degeneration** of ν and we write $\nu \to \mu$. The degeneration $\nu \to \mu$ is called **trivial** if μ is isomorphic to ν. A Jordan algebra is called **rigid** if it admits non-trivial deformations, equivalently, the closure of its orbit in Jor_n coincides with an irreducible component of Jor_n.

One of the main tool to determine rigid algebras

Theorem (Gerstenhaber)

If $H^2(\mu, \mu) = 0$ then μ is rigid.

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Geometric Invariants for $\mu, \nu \in Jor_n$ and $\mu \to \nu$

1. $\dim Aut(\mu) \leq \dim Aut(\nu)$
2. any polynomial identity T which is valid in μ is valid in ν
3. $\dim Rad(\mu) \leq \dim Rad(\nu)$
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Asymptotic of dimensions of Alg_n:

- Yu. Neretin showed that the dimension of $Assoc_n$ is at most $\frac{4}{27}n^3 + O(n^{\frac{8}{3}})$, while of Lie_n and $Comm_n$ is $\frac{2}{27}n^3 + O(n^{\frac{8}{3}})$.
- IK, I. Shestakov showed that the dimension of Alt_n is at most $\frac{4}{27}n^3 + O(n^{\frac{8}{3}})$, and conjectured that the dimension of Jor_n is $\frac{1}{6\sqrt{3}}n^3 + O(n^{\frac{8}{3}})$.

How to describe the open subspaces of Jor_n?

Let J_i^α, $i \in I$ be a family of algebras. Denote

$$\mathcal{N}_i^{\#} = \bigcup_{\alpha \in T} G \cdot (J_i^\alpha) \, .$$

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Infinite family is dominated by a single algebra

Example ($J_{22} \to \mathcal{N}_{15}^{\#}$, IK, Martin)

$$J_{22}: \quad \begin{array}{c} n_1^2 = n_2 \\ n_1 n_2 = n_3^2 = n_4 \end{array} \qquad\qquad J_{15}^{\alpha}: \quad \begin{array}{c} n_1^2 = n_2 \\ n_1 n_2 = n_3 \\ n_4^2 = \alpha n_5 \end{array}$$

For any $\alpha \in \mathrm{k}$ consider a change of basis of J_{22}

$$\gamma_\alpha(t): \quad \begin{array}{l} e_1^t = t^2 n_1 - \frac{\alpha}{3} n_2 + t n_3, \\ e_2^t = t^4 n_2 + (1 - \frac{2\alpha}{3}) t^2 n_4 + \frac{\alpha^2}{9} n_5, \\ e_3^t = t^6 n_4 - (-1 + \alpha) t^4 n_5, \\ e_4^t = t^4 n_2 - t^5 n_3 + \frac{\alpha t^2}{3} n_4, \\ e_5^t = t^8 n_5 \end{array} \qquad \begin{array}{c} \det(\gamma_\alpha) = t^{25} \\ \forall t \neq 0, \ \gamma_\alpha(t) \in J_{22}^G. \end{array}$$

Since $(e_4^t)^2 = t^4 e_3^t + \alpha e_5^t$ it follows that $\gamma_\alpha(0)$ gives the multiplication table of J_{15}^{α}.

Example ($J_{23}^{\beta} \to J_{41}$, IK,Martin)

$$J_{23}^{\beta}: \quad \begin{array}{c} n_1^2 = n_3, \;\; n_1 n_3 = n_2 n_4 = n_5 \\ n_1 n_2 = n_4, \;\; n_2 n_3 = \beta n_5 \end{array} \qquad J_{41}: \quad \begin{array}{c} n_1^2 = n_5, \;\; n_1 n_2 = n_3 \\ n_2^2 = n_5 \end{array}$$

No single member of J_{23}^{β} can dominate J_{41}, since $\dim Aut(J_{41}) = \dim Aut(J_{23}^{\beta})$.
Consider the change of basis of $\mathcal{J}_{23}^{\beta(t)}$, where $\beta(t) = \frac{1}{t}$

$$\gamma(t): \quad \begin{array}{c} e_1^t = tn_1 - \frac{1}{2} n_2, \quad e_2^t = -\frac{1}{2} n_2, \\ e_3^t = -\frac{t}{2} n_3, \quad e_4^t = \frac{t}{4} n_5, \quad e_5^t = t^2 n_3 - tn_4 \end{array}$$

So the curve $\gamma(t)$ lies transversely to the orbits of $\mathcal{J}_{23}^{\beta}$, meaning that, for any $t \neq 0$, $\gamma(t) \subset \mathcal{N}_{23}^{\#}$ and cuts J_{41}^{G} in $t = 0$. Thus $J_{41} \in \overline{\mathcal{N}_{23}^{\#}}$.

Back to Geometry: Kempf-Kirwan-Ness Theory

The **Kempf-Ness** theorem provides a beautiful and simple geometric criterion for the closedness of orbits of a rational representation of a complex reductive linear algebraic group. It says an orbit is closed iff it contains a minimal vector, i.e. a vector minimizing the distance to the origin. It also implies that a non-closed orbit with positive distance to the origin contains a non-trivial closed orbit in its closure.

On the other hand, the union of all the orbits containing the origin in its closure comprise the so-called **null cone**. The **Kirwain-Ness Theorem** describes a Morse-type **stratification** of the null cone into finitely many invariant subvarieties with respect to the energy functional associated to the moment map.

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Moment map

For G consider its (change of basis) action on V_n and its maximal compact subgroup $K := U(n)$. The Hermitean product on $\mathbb{C}^n$ canonically extends to a Hermitean product on V_n:

$$\langle \mu, \nu \rangle = \sum_{ij} \langle \mu(x_i, x_j), \nu(x_i, x_j) \rangle,$$

where $\mu, \nu \in V_n$ and $\{x_1, \ldots, x_n\}$ is an orthonormal basis of $\mathbb{C}$. Write $\mathfrak{g}$ and $\mathfrak{k}$ for the Lie algebras of Lie groups G and K, thus $\mathfrak{g} = \mathfrak{k} + i\mathfrak{k}$. The derived action by $\mathfrak{g}$ on V_n is given by

$$A \cdot \mu(x, y) = A(\mu(x, y)) - \mu(Ax, y) - \mu(x, Ay)$$

for $A \in \mathfrak{g}$, $\mu \in V_n$, $x, y \in \mathbb{C}^n$. We also consider the Ad_K invariant Hermitean product on $\mathfrak{g}$ given by $(A, B) = Tr(AB^*)$ for $A, B \in \mathfrak{g}$. The **scale-invariant moment map** $m : V_n \to i\mathfrak{k}$ is

$$(m(\mu), X) = \frac{1}{\|\mu\|^2} \langle X \cdot \mu, \mu \rangle, \qquad \mu \in V_n,\ X \in i\mathfrak{k}.$$

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Solitons

- m is K-invariant

$$m(k \cdot \mu) = km(\mu)k^{-1}$$

for all $\mu \in V_n$, $k \in K$.

- Since $\langle X \cdot \mu, \mu \rangle = \frac{1}{2}\frac{d}{dt}|_{t=0}\|e^{tX} \cdot \mu\|^2$, one sees that $m(\mu)$ controls the length of vectors near μ. One shows that the zeros of μ correspond to the minimal vectors.
- The **energy functional** $E_n : V_n \setminus 0 \to \mathbb{R}$ is the scale- and K-invariant functional given by

$$E_n(\mu) = \|m(\mu)\|^2.$$

Thus the zero set of E_n corresponds to the minimal vectors. The other critical points of E_n are called **solitons**. Kirwan and Ness showed that there are finitely many K-orbits of solitons, they all occur in the null cone N and determine a G-invariant stratification of $N \setminus \{0\}$ by locally closed irreducible nonsingular subvarieties. In our setting $N = V_n$.

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Distinguished orbits

We say $G \cdot \mu$ is **distinguished** if it contains a soliton (not every orbit is distinguished)

Theorem (Gorodski-IK-Martin)

1. *Every isomorphism class of complex semisimple Jordan algebra is distinguished. Further, in each dimension the semisimple Jordan solitons are characterized among Jordan algebra as having the lowest possible value of the energy for that dimension.*
2. *Every isomorphism class of complex Jordan algebra of dimension at most 4 is distinguished except the nilpotent algebra* $n_1 n_2 = n_3,\ n_1 n_3 = n_4,\ n_2^2 = n_4$.

Proposition (Gorodski-IK-Martin)

$G \cdot \mu$ is a distinguished orbit and the energy of a soliton in $G \cdot \mu$ is larger than $E_n(\nu)$ then μ cannot degenerate to ν.

We also obtain a simple cohomology free proof of the following well knows result:

Theorem (Glassman 1970, Finston 1990)

Every finite-dimensional complex semisimple Jordan algebra is rigid.

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Structure of solitons

Proposition (Gorodski-IK-Martin)

The moment map $m : V_n \to i\mathfrak{k}$ *is given by* $m(\mu) = \frac{1}{\|\mu\|^2} M_\mu$ *where* $M_\mu = -2\sum_{i=1}^n L_{x_i}^{\mu *} L_{x_i}^{\mu} + \sum_{i=1}^n L_{x_i}^{\mu} L_{x_i}^{\mu *}$ *and* $x_1, \ldots, x_n$ *is orthonormal basis of* $\mathbb{C}^n$.

Theorem (Gorodski-IK-Martin)

Let $\mu \in V_n$ *and write* $J = (\mathbb{C}^n, \mu)$ *for the corresponding commutative algebra. Then*

1. μ *is a soliton if and only if its moment matrix* M_μ *is given by* $M_\mu = c_\mu I + D_\mu$, *where* $c_\mu < 0$ *and* D_μ *is a derivation of* μ *and a Hermitian matrix; further, in this case there is a positive multiple of* D_μ *which has rational eigenvalues.*
2. *If* μ *is a Jordan soliton then there is a maximal semisimple subalgebra* J_s *of* J *such that* $J = J_s + R$ *is a* D_μ*-invariant decomposition, where* R *denotes the radical of* J. *Further* $D_\mu|_{J_s} = 0$.

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Open problems

For a Jordan soliton $J = (\mathbb{C}^n, \mu) = J_s + R$:

- Are the eigenvalues of the Hermitian derivation D_μ positive on the radical R?
- Is the radical R also a soliton?
- Conversely given nilpotent Jordan soliton R and semisimple Jordan algebra J_s, such that $J = J_s + R$ is Jordan algebra, can we extend the metric from R to J so that J becomes a soliton?

C. Gorodski, I. Kashuba, M. E. Martin, *A moment map for the variety of Jordan algebras*, Communication in Contemporary Mathematics, 27 (03), 2450015, 2025.

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Happy Anniversary, Claudio!